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Sketch retrieval and relevance feedback with biased SVM classification

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ABSTRACT

This paper proposes an effective approach for content-based sketch retrieval. It addresses three characteristics as follows. Firstly, both structural relations and global shape descriptors are combined to represent sketch content. Secondly, feature weighting and combination are performed to obtain a reasonable mechanism for similarity calculation. Finally, relevance feedback based on biased SVM (BSVM) algorithm is employed to capture user's query interests online and thus improve retrieval performance. Experiments prove the effectiveness of our proposed method in sketch retrieval.

1. Introduction

As a natural form of user interaction, sketching interfaces have recently attracted a lot of attention. They are expected to provide flexible, informal interaction between computers and users that do not hinder creative thinking. As sketch objects are being digitized, more and more digital sketches with useful information will be generated. There is an urgent demand for effective tools to facilitate the retrieval of sketches. The goal to find similar sketches from large collections or from remotely distributed databases is shared not only by researchers, educators and professionals, but also by general users.

Most existing sketch retrieval approaches rely on non-learning mechanism such as (Leung, 2003; Fonseca and Jorge, 2002; Heesch, 2001). These non-learning methods are limited by vulnerable parametric assumptions, the choice of appropriate thresholds, and the ad hoc approaches for representing sketch content. Moreover, computers can only extract low-level visual features that are computable, such as curvature, pen speed, and so on. But people's needs and understandings are high-level concepts, and they cannot acquire the exact low-level visual features as computers do. This makes computers unable to capture user's mind exactly. Meanwhile, user's interpretation of sketch changes under different circumstances or contexts. Given the same sketch, different users may have different ways of interpretation due to their individual knowledge background. Even the same person may have different understandings at different times or in different domains, which makes the problem more difficult. In order to bridge the gap between human and computers, sketch retrieval system must have the ability of real-time learning, which associates user's subjective high-level understandings to low-level computable features online. That is to say, we must shift towards more interactive mechanisms aimed at establishing more precisely the particular information need of users.

Relevance feedback is the most natural strategy to alleviate the above problems by introducing user judgments, and it has been shown as the most powerful and promising tool to improve the performance of retrieval systems (Huang and Zhou, 2001). The idea is to let user label each of the retrieved items as whether it is relevant or not. This information is fed back and used by the system to refine searching strategies online to get better results. The concept of relevance feedback was first used in text retrieval (Salton, 1989), but it come to better usage in CBIR, and has been developed a lot from early heuristic weight adjustment to optimal learning methods (Huang and Zhou, 2001). Same as CBIR, introducing relevance feedback into sketch retrieval could also improve retrieval performance to a large extent, since it is difficult to establish the accurate mapping between query concepts and feature representation without user interaction. Some sketch retrieval systems have made primary attempts (Leung, 2003; Heesch, 2001), but the feedback strategy they used is only the easiest method of query point moving and results are not satisfactory. We therefore decide to employ learning techniques to further refine the retrieval performance, and thus bring computer perception closer towards human mind.

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However, relevance feedback in sketch retrieval differs from that of CBIR (Veltkamp and Tanase, 2000) mainly in two aspects. The first is the problem of sketch content representation. Unlike images in CBIR (Faloutsos et al., 1994), sketches in electronic format are stored in structured form instead of basic bitmap format, in which each basic unit is a stroke rather than a pixel. Moreover, while color, texture and shape features can be extracted from natural images, only shape information is available in sketches. Accordingly, sketches contain more structural and semantic information and greater care needs be taken when choosing sketch representations for the one and only feature available. Second, relevance feedback in sketch retrieval should be more capable than that in CBIR (Sciascio and Mongiello, 1999). This is because, with shape information being the principal or only feature used to describe content, it is arguably even harder for computers to capture user's interests in sketch retrieval. Based only on shape feature, user's intentions may be even ambiguous or changeable. Inferring user's interests will be accordingly much more complicated. Therefore, relevance feedback in sketch retrieval requires more effective learning techniques to accomplish the retrieval task.

In this paper, we employ both structural and global features to represent sketch content and introduce learning techniques based on biased SVM (BSVM) during relevance feedback into sketch retrieval systems, in which user's subjective perception is learned after candidate results are generated by initial similarity matching procedure.

The rest of the paper is organized as follows. In Section 2, we first discuss features extracted for sketch content representation and corresponding similarity matching to obtain candidate results. In Section 3, we analysis the property of relevance feedback and present our feedback process in sketch retrieval with learning techniques based on biased SVM. Experiments, performance evaluations, and discussion are given in Section 4. Finally, Section 5 concludes our work.

2. Sketch content representation

For sketch retrieval, content representation is an important step, since the more effective the features are extracted to represent sketch content, the more accurate the sketches are matched, which will directly influence the overall performance of retrieval system and the following relevance feedback procedure.

As sketch is usually informal and ambiguous, not accurate, the content what sketch retrieval stressed would be its graphical features. In order to represent sketch content comprehensively, we extract both structural and global information and construct corresponding feature vectors. Structural representation which views sketch as a set of primitives is essential, since sketch has intrinsic structural characteristics. On the other hand, global representation reflects sketch in a whole and is close to people's visual perception. By combining both kinds of information, we can describe sketch content more comprehensively and effectively than using only one of them.

2.1. Preprocessing

There may be a great diversity of original sketch data under user variations, which must be preprocessed first. We perform stroke segmentation based on pen speed and curvature information (Sezgin, 2001) to decompose sketches of both user's inputting query pattern and that in stroke-based sketch library into geometric primitive shapes as lines, arcs and ellipses. During drawing process, we monitor user's activity and split strokes at locus of where pen speed reaches the minimal value or where variety of curvatures occurs evidently. Since firstly, there are too many kinds of strokes and it is impossible to define all of them; and secondly, a sketchy graphics can be organized by different strokes in different manners, but the same graphics can only be composed of a group of basic primitive shapes (lines, arcs and ellipses) in a fixed way, stroke segmentation can therefore create a relevant consistent representation of sketches and low down retrieval complexity to a large extent.

2.2. Structural features

In our previous work (Liang et al., 2005), we extracted structural features of sketches. These features are composed of eight kinds of spatial relations between primitives among constitutes of sketch, including cross, half-cross, adjacency, parallelism, cut, tangency, embody and ellipse–ellipse intersection, as shown in Fig. 1.

After a sketch is segmented into primitives, we will calculate these spatial relations among different primitives. For each kind of spatial relation, we collect several corresponding pairs of primitives. We then represent these spatial relations in the form of topology graphs. Let $G = (V,E)$ be a topology graph representing a kind of spatial relation, where V is the set of graph vertices and E is the set of graph edges. Every vertex in G corresponds to a constitute unit, or a primitive shape of sketch. Edge $(v_i, v_j) \in E$ if and only if certain spatial relation exists between the two vertices/primitive shapes v_i and v_j. In this way, the matching of sketch spatial relations will be transformed to the problem of matching between their topology graphs.

In order to avoid the complexity of graph matching, we employ Laplacian spectrum to describe topology graph. Graph spectrum is the set of graph eigenvalues, and we can calculate the spectrum of a graph from the eigenvalues of its Laplacian matrix. Laplacian spectrum is a stable prosperity of graph, as well as an effective way to map topology graphs into corresponding multidimensional vectors (Shokoufandeh et al., 1999). Moreover, Laplacian spectrum has more representational power than adjacency spectrum, which was originally used in our previous work (Liang et al., 2005), in terms of generating much less cospectral possibilities (Haemers and Spence, 2004). If we let $A(G)$ be the adjacency matrix of G whose element at position (i,j) is

$$A(i,j) = \begin{cases} 1 & \text{if } (v_i, v_j) \in E, \\ 0 & \text{otherwise}. \end{cases} \tag{1}$$

Let $D(G)$ be the diagonal matrix of vertex degrees with elements of index (i,i) as

$$D(i,i) = \sum_{v_j \in V} A(i,j). \tag{2}$$

Then the Laplacian matrix $L(G) = D(G) - A(G)$ is calculated as follows:

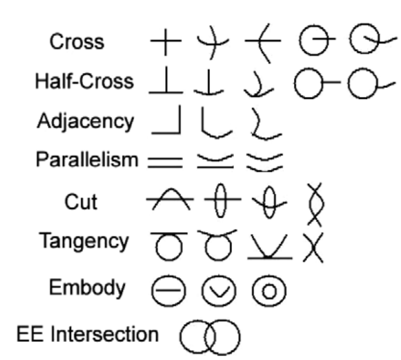

Fig. 1. Classes of primitive relations.

$$L(i,j) = \begin{cases} \text{degree}(v_i) & \text{if } i = j, \\ -1 & \text{if } i \neq j \text{ and } (v_i, v_j) \in E, \\ 0 & \text{otherwise}. \end{cases} \tag{3}$$

The Laplacian spectrum of graph G is then obtained from the eigendecomposition of its Laplacian matrix $L(G)$. And these eigenvalues together form the spectrum vector to describe the graph. In this manner, the problem of graph matching could be reduced to comparison of spectrum vectors.

Moreover, the dimension of spectrum vectors is equal to the number of vertices in topology graph, which depends on the number of primitive shapes in original sketch data. So sketches composed of different number of primitives will result in spectrum vectors with different dimensions. In order to compare sketches with different primitives, all spectrums are normalized to their L2 norms. Thus we reduce the n-dimensional spectrum vector $d = [d_1, d_2, \ldots, d_n]$ to the Euclidean distance:

$$\|d\| = \sqrt{d_1^2 + d_2^2 + \cdots + d_n^2}. \tag{4}$$

When applying to all the eight classes of spatial relations, we will get an 8-dimensional feature vector $\boldsymbol{x}$, in which the ith dimension represents the topology of the ith spatial relation between sketch primitives.

Finally, these structural features need to be normalized with all sketches in database. The normalization is carried out according to the following equation, which is simple and effective, at the meantime keeps the order of features within the dimension.

$$\boldsymbol{x}_i = \frac{\boldsymbol{x}_i}{\max\limits_{\boldsymbol{x}_i \in D_i}(\boldsymbol{x}_i)}, \quad 1 \leqslant i \leqslant 8, \tag{5}$$

where $\max_{\boldsymbol{x}_i \in D_i}(\boldsymbol{x}_i)$ is the maximal feature value among all database sketches in the ith feature dimension D_i.

2.3. Global features

In addition to structural features, we consider seven types of global descriptors in our sketch retrieval system. The first one is sketch eccentricity, which equals to the ratio of length of major axis to length of minor axis of the sketch. The second and third features are normalized distances of sketch centroid in major and minor axis orientations, which reflect the sketch centroid's position in its minimum enclosing rectangle (MER). In case the centroid is not in the center, the distances to the opposite edges of MER might be different and we will take the shorter one for calculation. The fourth feature is the average distance between centroids of sketch primitives, normalized by the maximal distance between any two primitive centroids to account for scaling. This descriptor gives information similar to aspect ratio of the sketch, but it is insensitive to rotation. The final three features form primitive histogram which is made up by the number of primitives of each type as line, arc, and ellipse, normalized by the total number of primitives. This histogram feature takes into account the primitive attribute information which is neglected by spatial relation calculation. For example, when extracting "Cross" spatial relation, two crossed lines and two crossed arcs will result in the same topology graph and thus the same spectrum. By means of considering primitive types, the histogram increases the representational power of sketch features. Finally, these descriptors together form a 7-dimensional global numeric vector.

Fig. 2 gives an example of computing sketch global features. Strokes of the sketch are first segmented into eight primitives of six lines and two ellipses in preprocessing. The rectangle is the MER of the sketch, L_{major} and L_{minor} are lengths of major and minor axis respectively, C is the centroid of the sketch, R_{major} and R_{minor} are distances of sketch centroid in major and minor axis orientations, p_i is the centroid of sketch primitive i ($i = 1,2,\ldots,8$). Then global features of the example sketch can be calculated as follows: eccentricity = L_{minor}/L_{major} = 0.700472; normalized distance of sketch centroid in major axis orientation = R_{major}/L_{major} = 0.452417; normalized distance of sketch centroid in minor axis orientation = R_{minor}/L_{minor} = 0.470988; average distance between primitive centroids = Avg[dis(p_i,p_j)]/Max[dis(p_i,p_j)] = 0.525816 ($i,j = 1,2,\ldots,8$); primitive histogram of line, arc and ellipse = Number of lines, arcs and ellipses/Number of primitives = 0.75, 0 and 0.25, respectively.

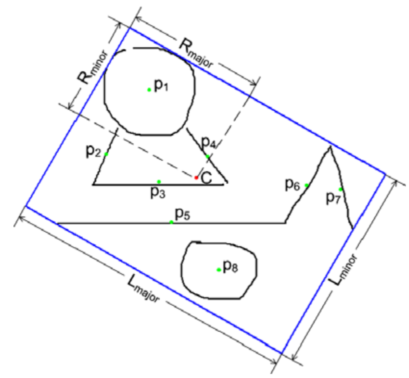

Fig. 2. Illustration of computing global features.

2.4. Feature combination and weighting

Structural and global features together can be organized to form a 15-dimensional feature vector, which accommodate scale, translation and rotation invariance. However, the importance or effect of each feature dimension in similarity matching is different. Feature weighting and selection becomes an important issue when aiming to find out an optimal feature combination pattern that best describes sketch content. We calculate weights of each feature dimension based on the whole database to increase the effect of feature dimensions which are more important. The main idea is that the better the feature component classifies the database, the more effective the component is and thus deserves a higher weight. For an effective feature component or dimension, the feature values should be similar to one another within the same sketch class and dissimilar to values in other classes. Therefore, the higher inter-class variance and lower intra-class variance the feature component has, the better it is to distinguish different sketch categories. A higher ratio of inter-class to intra-class variance will indicate a better feature component for similarity matching.

Assume that the database constitutes of c classes of sketches as $\{\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_c\}$. $\boldsymbol{x}$ is a sample sketch in the database. The number of sketches in class k is n_k, and the total number of sketches in the database is n. Then the intra-class variance could be calculated as follows:

$$\textbf{Variance}_{\text{intra}} = \frac{1}{n}\sum_{k=1}^{c}\sum_{\boldsymbol{x}\in\boldsymbol{X}_k}(\boldsymbol{x}-\boldsymbol{\mu}_k)(\boldsymbol{x}-\boldsymbol{\mu}_k)^{\mathrm{T}} \tag{6}$$

in which $\boldsymbol{\mu}_k = \frac{1}{n_k}\sum_{\boldsymbol{x}\in\boldsymbol{X}_k}\boldsymbol{x}$ is the mean value of class k. The inter-class variance could then be formulated as

$$\textbf{Variance}_{\text{inter}} = \frac{1}{n}\sum_{k=1}^{c} n_k(\boldsymbol{\mu}-\boldsymbol{\mu}_k)(\boldsymbol{\mu}-\boldsymbol{\mu}_k)^{\mathrm{T}}, \tag{7}$$

where $\boldsymbol{\mu} = \frac{1}{n}\sum\boldsymbol{x}$ is the mean feature value among all classes. Then the evaluation of the weight vector will be set according to the ratio of inter-variance to intra-variance of a certain dimension, which is

$$\boldsymbol{w}_i = \frac{\textbf{Variance}_{\text{inter}}(i,i)}{\textbf{Variance}_{\text{intra}}(i,i)}. \tag{8}$$

Then the weights will be normalized as follows, so that the sum of all dimensions is to be 1:

$$\boldsymbol{w} = \frac{\boldsymbol{w}}{\sum_i \boldsymbol{w}_i}. \tag{9}$$

By incorporating weight values into the following matching procedure of sketch retrieval, we can augment the importance of valuable feature components and thus obtain an optimal feature representation pattern of sketch similarity calculation.

2.5. Content-based similarity matching

Similarity matching is performed between feature vectors of compared sketches based on weighted Euclidean distance as

$$\text{dis}(\boldsymbol{x},\boldsymbol{x}') = \sqrt{\sum_{i=1}^{15}[\boldsymbol{w}_i \times (\boldsymbol{x}'_i - \boldsymbol{x}_i)^2]}. \tag{10}$$

The accordingly similarity between sketches can be calculated derived from this weighted Euclidean distance as follows:

$$\text{similarity}(\boldsymbol{x},\boldsymbol{x}') = \begin{cases} 0 & \text{if dis}(\boldsymbol{x},\boldsymbol{x}') \geqslant \delta, \\ 1 - \frac{\text{dis}(\boldsymbol{x},\boldsymbol{x}')}{\delta} & \text{otherwise}. \end{cases} \tag{11}$$

where δ is a threshold introduced from our experiments to normalize similarity and make it fall into the range of $[0, 1]$. Sketches with the highest similarities in database are extracted as candidate result set and returned to users on interface.

3. Relevance feedback based on biased SVM

Relevance feedback in sketch retrieval is treated as a biased learning or classification problem (Huang et al., 2002) in this paper. After candidate results are given by means of low-level similarity matching, user labels the results online to indicate positive (relevant) or negative (irrelevant) ones, which will be regarded as training samples for the classifier. By learning this feedback information, the classifier will analyze the properties of positive and negative data, and try to estimate the distribution of user's target sketches in feature space. Retrieval system then employs the trained classifier to distinguish the whole feature base into two classes of relevant and irrelevant sketches, and return the most relevant sketches to user as the refined results.

3.1. A new point of view for relevance feedback: biased classification

Typical relevance feedback approaches with learning techniques are based on strict binary classification, which treats positive and negative instances equally (Zhou and Huang, 2001). However, binary classification does not consider the imbalance problem in relevance feedback, which is the number of negative instances are significantly larger than positive ones. This imbalanced dataset problem will lead to the positive data be overwhelmed by negative data. Moreover, though it is reasonable to assume positive instances to cluster in a certain manner, either linearly or non-linearly, negative instances do not cluster since they usually could come from any different class. It is almost impossible to estimate the real distribution of negative instances in the database based only on the negative information that user labeled. Forcefully assigning all negative instances into one class or mode will mislead the algorithm and decrease the performance, especially in the case of relevance feedback where the number of training samples is small.

This kind of situation is regarded as a "biased classification problem" (Huang et al., 2002), in which user only cares about one certain class among several classes. Or in other words, user is biased toward one class but neglect other ones, and it is desirable to distinguish a real binary problem from this biased classification. The mathematical definition of "biased classification" is stated as follows:

Definition (***Biased classification***). Assume the sample space Ω constitutes of n classes with label $\{\omega_1, \omega_2, \ldots, \omega_n\}$, x is a sample in Ω, then biased classification is to find out a discriminant function $BC(x)$, such that

$$\forall x \in \Omega, \quad BC(x) = \begin{cases} 1 & \text{if } x \in \omega_i, \\ -1 & \text{if } x \notin \omega_i, \end{cases} \tag{12}$$

where ω_i is the class that the user is interested in.

In relevance feedback, training instances of sketches are labeled as positive and negative only according to whether they belong to the target class or not. Thus negative instances come from any unconcerned class, and the small number of training examples cannot represent the accurate distribution of all the negative samples in feature space. Therefore, the problem we meet in typical relevance feedback is in fact an exactly biased classification problem.

3.2. Relevance feedback with SVMs

Recently, researchers introduce SVM into relevance feedback to perform feature selection or dataset classification. Since SVM has good generalization ability, SVM-based techniques are considered as the most promising ones when implementing relevance feedback (Qian et al., 2004). Typical SVM relevance feedback approaches are based on binary classification (Qian et al., 2004) or one-class classification (Chen et al., 2001). However, as we mentioned above, binary SVM places equal treatments to both positive and negative samples and neglects the imbalance data problem. So it cannot fulfill the needs of biased classification and does not suit for modeling relevance feedback. One-class SVM (one SVM, 1SVM) (Schölkopf et al., 2001) seems to avoid the imbalance problem. It regards relevance feedback as estimating the distribution of target items in feature space, and fits a tight hypersphere to include most of the target items based on positive examples. However, 1SVM converges quickly and cannot reach the optimal performance without the help of negative information.

In order to attack the biased classification problem and make use of both positive and negative information, we introduce learning techniques based on BSVM (Hoi et al., 2004) into sketch retrieval. The distribution of target sketches is estimated in feature space on the promise of not violating user's feedback information.

3.3. Biased support vector machine

The main idea of BSVM associates principles of both binary SVM and 1SVM. But unlike SVM's using a pair of parallel hyperplanes and 1SVM's using a single hypersphere to describe the data, BSVM employs a pair of concentric hyperspheres to accomplish it. The inner hypersphere is set as small as possible while it includes most of the positive data. On the other hand, the outer hypersphere tries to be as large as possible while it contains least of the negative data. In other words, we want the inner hypersphere capture most of the positive while the outer one push out the negative. Therefore, the goal of our biased classification is to find an optimal hypersphere, which can include most of the positive instances while exclude most of the negative data with the help of user's labeled information. Fig. 3 illustrates the hyperspheres of BSVM in a 2D feature space example. The solids and crosses denote for relevant and irrel-

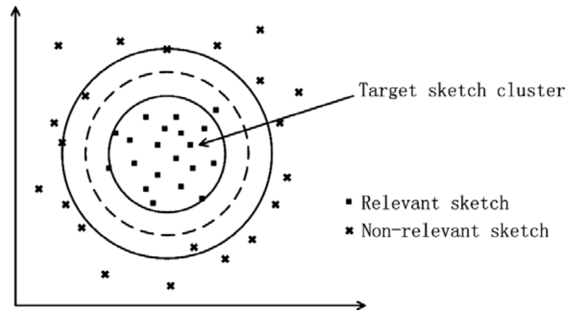

Fig. 3. Illustrations of BSVM hypersphere in 2D feature space.

evant sketches respectively, while the dash line indicates the objective optimal hypersphere for decision.

The task of obtaining the optimal hypersphere of BSVM can be formulated as an optimization problem. Let us consider the training data set as

$$(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n) \in \Re^d \times Y, \quad Y = \{-1, +1\}, \tag{13}$$

where $n \in N$ is the number of training samples, $d \in N$ is the dimension of feature space, $\boldsymbol{x}_i$ is a feature vector and y_i is its class label of the training data which equals to 1 if the sketch is relevant and -1 if it is irrelevant.

The primal form of the objective function for the optimal hypersphere is shown in the following form (Hoi et al., 2004):

$$\begin{aligned} \min_{R\in\Re,\xi\in\Re,\rho\in\Re,c\in F} \quad & R^2 - v\rho + \frac{1}{n}\sum_{i=1}^{n}\xi_i \\ \text{s.t.} \quad & y_i(\|\Phi(\boldsymbol{x}_i) - \mathbf{c}\|^2 - R^2) \leqslant -\rho + \xi_i, \\ & \rho \geqslant 0,\ 0 \leqslant v \leqslant 1,\ \xi_i \geqslant 0,\ i = 1, \ldots, n, \end{aligned} \tag{14}$$

where ξ_i are slack variables for margin errors, and $\xi = 0$ indicates linear case in which training data can be separated, otherwise the case is non-linear. $\Phi(\boldsymbol{x}_i)$ is mapping function, $\mathbf{c}$ and R are center and radius of the optimal hypersphere, ρ is width of the margin. Parameter v can be adjusted when running the algorithm to trade off between the number of support vectors and margin errors.

By introducing Lagrange Multipliers, the optimization problem can be shown in dual form with kernel functions as (Hoi et al., 2004)

$$\begin{aligned} & \min_{\alpha} L = \min_{\alpha} \sum_{i,j=1}^{n} \alpha_i\alpha_j y_i y_j k(\boldsymbol{x}_i, \boldsymbol{x}_j) - \sum_{i=1}^{n} \alpha_i y_i k(\boldsymbol{x}_i, \boldsymbol{x}_i) \\ & \text{s.t.} \quad 0 \leqslant \alpha_i \leqslant \frac{1}{n}, \quad \sum_i \alpha_i y = 1, \quad \sum_i \alpha_i \geqslant v, \quad i = 1, \ldots, n. \end{aligned} \tag{15}$$

The optimal α_i can be obtained after solving this dual problem by quadratic programming (QP) techniques. Then the resulting decision function can be defined as (Hoi et al., 2004)

$$\begin{aligned} f(\boldsymbol{x}) &= \mathrm{sgn}(R^2 - \|\Phi(\boldsymbol{x}) - \mathbf{c}\|^2) \\ &= \mathrm{sgn}\left[-k(\boldsymbol{x},\boldsymbol{x}) + 2\sum_{i=1}^{n}\alpha_i y_i k(\boldsymbol{x}_i,\boldsymbol{x}) + R^2 - \sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j k(\boldsymbol{x}_i,\boldsymbol{x}_j)\right]. \end{aligned} \tag{16}$$

3.4. BSVM feedback in sketch retrieval

Sketches can be categorized into relevant and irrelevant based on the above decision function. Data point that lie inside the hypersphere ($f(\boldsymbol{x}) > 0$) will be predicted as positive, otherwise negative. Since both R and $\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j k(\boldsymbol{x}_i, \boldsymbol{x}_j)$ are same to all sketches, these constant values can be eliminated in relevance feedback where evaluation of sketch relevance is taken. Therefore we take a concise form of the decision function to be evaluation function:

$$e(\boldsymbol{x}) = -k(\boldsymbol{x}, \boldsymbol{x}) + 2\sum_{i=1}^{n}\alpha_i y_i k(\boldsymbol{x}_i, \boldsymbol{x}). \tag{17}$$

We will use it to represent sketch relevance in this paper.

Once parameters α_i are solved, evaluation function can be constructed and relevance of the sketch can be computed. Consequently, sketches in database are all ranked based on their relevance obtained from $e(\boldsymbol{x})$. Since sketches with the highest relevance are most likely to be the target ones which user is interested in, they are regarded as feedback results and returned to user.

From geometric perspective, BSVM tries to describe data and estimate their cluster area in feature space by constructing a pair of concentric hyperspheres, thus distinguish user's target sketches from other irrelevant ones. Moreover, from the results of mathematic deduction in optimization function, BSVM has the following constraint in its dual form (Hoi et al., 2004):

$$\sum_{i\in\mathrm{Rel}}\alpha_i - \sum_{i\in\mathrm{Irrel}}\alpha_i = 1, \tag{18}$$

where Rel and Irrel denote relevant and irrelevant sketch class respectively. This shows that the importance or weight allocated to positive instances in BSVM will be larger than negative ones, which means users are biased to the relevant class. Therefore, BSVM is more suitable for modeling the imbalance problem in relevance feedback of sketch retrieval system, while binary SVM is not effective enough to solve it.

4. Experiments and evaluation

In order to validate our proposed approach for learning in sketch retrieval, we carried out experiments in the environment of Pentium 4 1.6G CPU, 512MB memory, Windows XP, Visual C++ 6.0. For data collection, we chose 30 classes of parts from refrigerator drawings and 25 symbols from electric circuit diagram, and establish a database with 55 classes of sketches, as shown in Fig. 4. Ten persons in our group draw two repetitions of each of 55 classes of components habitually. Therefore a database with a total of 1100 sketches is obtained.

4.1. Sketch retrieval prototype

The process of our sketch retrieval approach with relevance feedback is shown in Fig. 5, which employs relevance feedback with biased SVM algorithm to refine the performance on the basis of initial similarity matching results.

Sketch retrieval system first computes candidate results after user have submitted a query sketch. Using sketch content representing approach we proposed in Section 2, features of both input

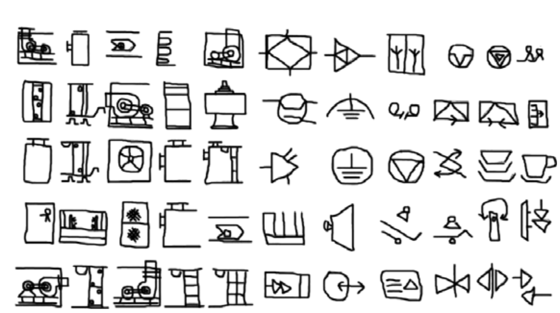

Fig. 4. Illustration of 55 classes of sketches.

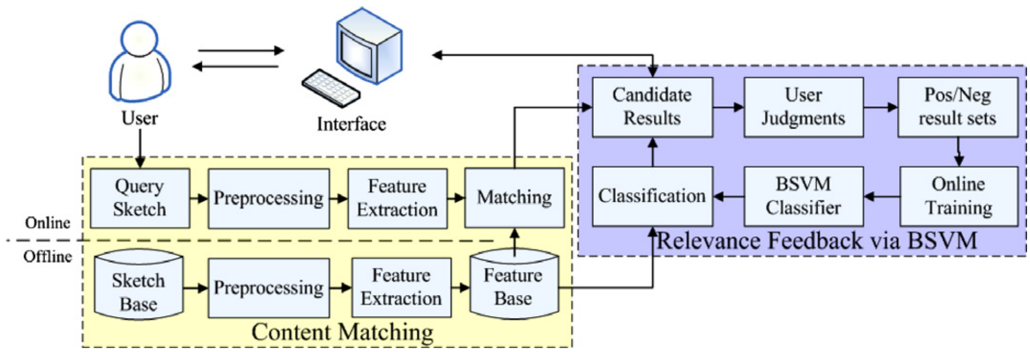

Fig. 5. Framework of sketch retrieval.

pattern of retrieval query and sketches in stroke-based sketch library are extracted online and offline respectively, and represented in the form of vectors. Content-based similarity matching is performed between feature vectors. Sketches with the highest similarities in database are extracted as candidate result set and returned to user on interface.

After candidate result set is generated, relevance feedback is used to refine the results in terms of user judgments. If the user has found out enough satisfied results which fall in the same class of the query sketch, the process of retrieval is over. Otherwise, the user marks the sketches in candidate result set and the system will start calling relevance feedback procedure, which Figures out the most relevant sketches by BSVM learning technique and returns them to user. The procedure of relevance feedback will be recalled, in which newly labeled instances will be used to train the classifier incrementally. The more times of relevance feedback, the closer the candidate results move to user's query intentions. Users may stop refining once they have retrieved enough similar sketches and quit the retrieval system.

4.2. Performance evaluation

To evaluate retrieval performance, recall and precision (RP) graph is plotted based on ranks of those sketches from the same class as query sketch. Recall is defined as the ratio between the number of relevant retrieved sketch and the total number of relevant sketches, and is calculated as

$$\text{recall} = \frac{|\text{relevant} \cap \text{candidates}|}{|\text{relevant}|} \tag{19}$$

Precision is defined as the ratio between the number of relevant retrieved sketches and the total number of retrieved sketches, and is calculated as

$$\text{precision} = \frac{|\text{relevant} \cap \text{candidates}|}{|\text{candidates}|}. \tag{20}$$

Obviously, when more items are returned, recall will be increased but precision will be decreased. In each of our experiments, three users each draw 55 query sketches of different classes to retrieve other sketches within the same class, which will be considered as positive results; otherwise negative. Retrieval results are averaged over all classes and users, and resulting graph is obtained by plotting these recall–precision pairs. In final RP graph, the higher the curve is, the better the retrieval performance is; and vice versa. Since for the same recall value, a higher curve signifies a higher precision value.

4.3. Experimental results and analysis

First, we will present the effectiveness of our structural and global features. Fig. 6 is RP curve of structural features extracted from Laplacian spectrum and adjacency spectrum. As we observe, Laplacian spectrum achieves better result than adjacency spectrum. This indicates that Laplacian spectrum has more representational power and can distinguish different classes more accurately. Fig. 7 shows the performance of global features and a subset of these features by removing the primitive histogram features. We could see that the combination of global features with primitive histogram performs better than that without it. This means employing global geometric descriptors together with primitive histogram, which takes into account the primitive attribute information that spatial relations calculation neglect, is effective for similarity matching.

Then we will show the effectiveness of our sketch content representation approach. Fig. 8 depicts RP curves of different features used in similarity matching. The four curves represent for performance of matching by global feature only, structural feature only, combination of both global and structural features, and weighted combination of these features. As we observe that, employing a combination of features from both structural and global aspects, we can describe the sketch content more effectively than using only one of them. After feature weighting and selection scheme figures out valuable and effective features for similarity matching, feature weights are assigned according to their importance to obtain an optimal feature combination pattern. As we can see in the graph, weighted feature representation achieves the best performance for similarity matching and is therefore more suitable for describing sketch content in retrieval system.

Next, we will evaluate the ability of BSVM learning approach for sketch retrieval when confronting the imbalance dataset problem in relevance feedback, or so called the biased classification problem. In our experiments, BSVM strategy is compared with three

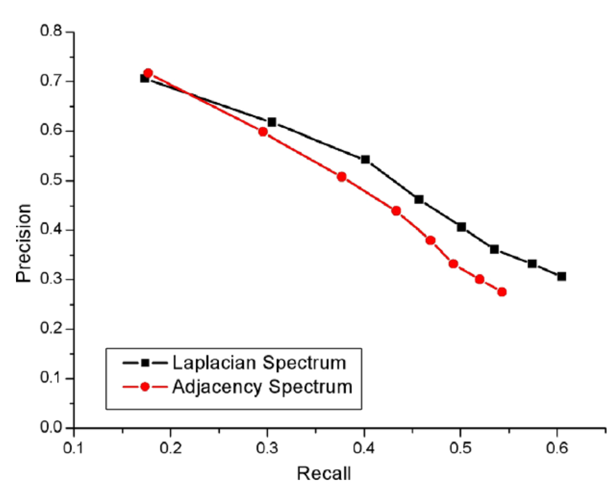

Fig. 6. Effectiveness of Laplacian spectrum.

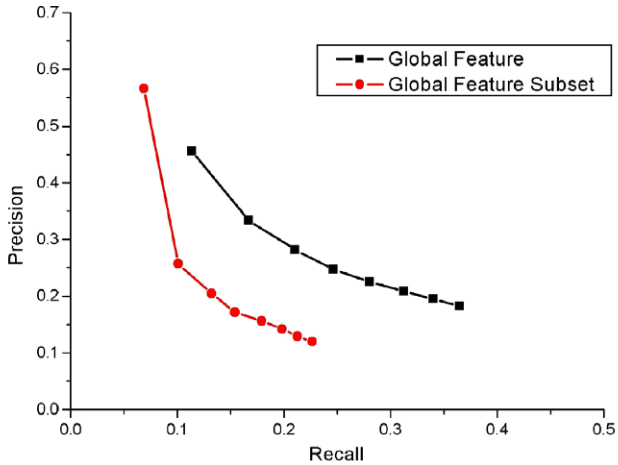

Fig. 7. Effectiveness of global features.

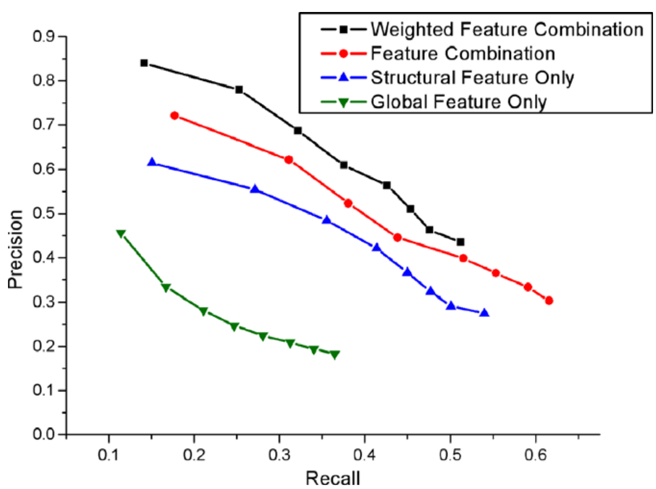

Fig. 8. Similarity matching of sketch retrieval.

other relevance feedback algorithms to show its capability. All these feedback algorithms are performed on the same basis of our content representation and matching approach. The first two feedbacks are SVM-based methods including binary SVM and 1SVM. Libsvm library (Chang and Lin, 2001) is introduced to develop these SVM-based algorithms in our experiment. Since experimental settings are important to impact on evaluation results, the same kernel (Radial Basis Function, RBF) and parameters ($\nu = 0.2$) are chosen for all SVM settings to enable an objective measure of performance without bias. Besides, we also compare BSVM feedback with a relevance feedback algorithm based on linear programming (LP) technique, which was incorporated into sketch retrieval by Li et al. (2006). In Li's research, he demonstrated that LP feedback can yield good results especially for its ability of dealing with the small sample problem of relevance feedback.

Fig. 9 shows RP graph of sketch retrieval with feedbacks. The five curves indicate retrieval performance of initial content-based sketch matching and that of relevance feedback with LP, SVM, 1SVM and BSVM, respectively. From the experimental results, we see that retrieval performance is refined noticeably by relevance feedbacks after initial similarity matching. The curve of BSVM feedback is the highest, which indicates BSVM feedback can improve the performance to a larger extent than the other three methods.

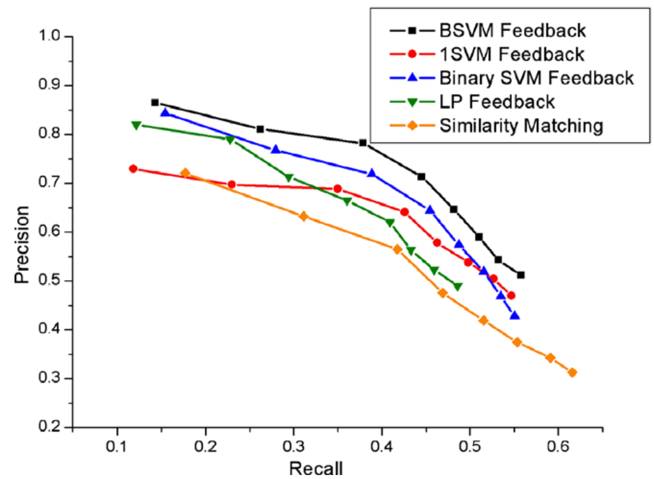

Fig. 9. Comparison of relevance feedbacks for sketch retrieval.

Or in other words, BSVM feedback can provide dramatic performance boost and thus retrieved sketches more accurately.

In order to evaluate performance of BSVM feedback under imbalance dataset condition, we randomly construct a database of $m + 1$ categories of sketchy symbols, which contain 1 positive class and m negative classes. In our experiment, we use $m = 1, 3, 5, 10, 15$, and obtain corresponding results. Fig. 10 is the relationship between average precision and the number of the negative classes in database with three rounds of feedback iterations. From the four curves we can see that, with the increasing of negative classes, performance of all feedbacks drop slowly. This is because when there are more negative classes, the imbalance dataset problem become more serious and retrieval precision will decrease. But as we can see in this graph, after three feedback iterations, the performance of BSVM feedback changes little and the curve of BSVM feedback nearly reaches horizontal, which indicates that BSVM is least influenced by data imbalance and can handle the asymmetric dataset very well. This is because BSVM aims to model biased classification problem to find the user wanted objects. Meanwhile, BSVM feedback has the highest precision values, which also indicates it has better performance and is more suitable to deal with the imbalance dataset problem in relevance feedback.

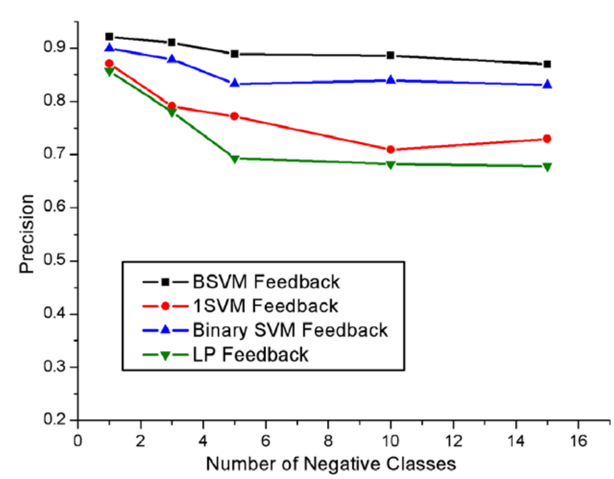

Fig. 10. Relationship between precision and number of negative classes.

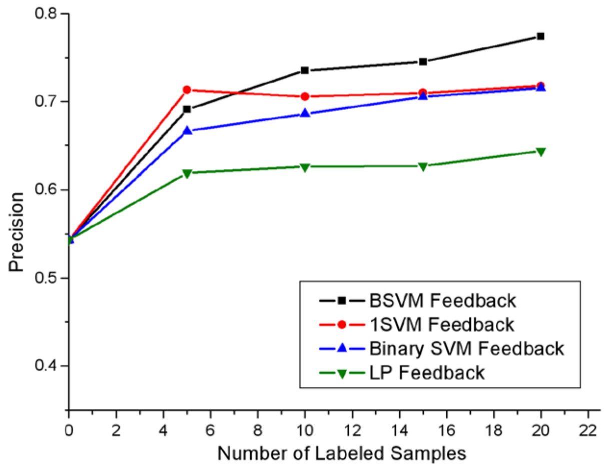

Fig. 11. Relationship between precision and number of labeled samples.

We also carry out experiment to evaluate feedback performance under the small sample case, in which users are asked to label different number of samples as 5, 10, 15 and 20, respectively, including both positive and negative samples. Fig. 11 shows the change of average precision with the number of labeled samples of feedbacks after one round of iteration. As we can observe that, with the increasing of training samples, all feedbacks refine the performance gradually. This is because the more training data that user provides, the more stable and meaningful results the classifiers will achieve. BSVM feedback yields the best precision figures and outperforms other approaches, which indicates BSVM learning technique works best in small sample training case of relevance feedback, and thus can improve retrieval precision more effectively. Binary SVM approach without considering the bias in retrieval task is not effective enough in addressing the relevance feedback problem. Moreover, we notice that the performance of 1SVM feedback in the beginning iterations is better than that of other approaches, but remains no big improvement in subsequent steps. This is because 1SVM can reach and estimate the enclosed positive region quickly, but it cannot be further improved without the help of the negative information, which results in the quick convergence of 1SVM feedback performance. While LP feedback only increases the performance to half of that of BSVM feedback when facing the same number of labeled samples. For example, with 5 labeled samples, BSVM feedback improves the performance with 17.1%, while LP feedback only improves the precision with 7.5%. These lead to a conclusion that when confronting with same number of labels, BSVM learning method can achieve better results than the other techniques and solve the relevance feedback problem well.

Finally, the time cost used in retrieval process is also a much-concerned factor for evaluating our approach's performance. Especially, in real-time interactive sketch retrieval environment, the time cost should be as small as possible. In our experiments, the average response time is about 87.2 ms for similarity matching and 150.7 ms for feedback. Generally, the time cost is much less than a second. The performance is sufficiently fast for real-time interaction.

5. Conclusion

Advances in Ubiquitous Computing and Ambient Intelligence have been going to boost the development and application of natural interfaces, and sketch retrieval would become one of the most important issues in multimedia retrieval. In this paper, we propose an effective approach for sketch retrieval with learning technique, which makes use of structural and global features to perform sketch content matching and incorporates BSVM into relevance feedback to refine retrieval performance on the basis of initial similarity matching results. Experiments demonstrate that the proposed content-based similarity matching is effective and learning via BSVM in relevance feedback is promising for improving retrieval performance and thus is suitable to learn user's query intention in sketch retrieval.

However, learning in sketch retrieval is a new and challenge issue. We have only made a primary attempt in this problem. Since labeled training set is usually a small subset of in multimedia database, unlabeled working sets are important clues for improving retrieval performance in sketch retrieval. We may consider partial labels and semi-supervised learning techniques and combine supervised and unsupervised learning algorithms to attack the relevance feedback problem in sketch retrieval. These guide the directions of our future work, in which we should consider using additional information to improve the precision of retrieval process.

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